

SummarAI

Individual Reflection

Jo

NLP Group Project · 2025–2026

Individual Reflection — Jo

My Contributions

Supabase Database Layer

I designed and set up the entire persistence layer for the app. I wrote `supabase_schema.sql`, which defines two tables: `projects` (for grouping meetings) and `summarizations` (for storing the full structured output). A key design decision was storing `key_takeaways` and `action_items` as JSONB columns rather than flat text, preserving the structured data as queryable fields rather than serialised strings.

Cascade delete: The schema includes a foreign key from `summarizations` to `projects` with cascade delete, so removing a project automatically removes all its meetings.

Row-level security: RLS policies allow public read/write access without requiring user accounts — appropriate for a POC and documented as a limitation for production.

I also verified that the backend endpoints matched what Supabase expected — checking that REST calls for creating, listing, and deleting projects and summarizations all worked correctly against the live database.

Field Review

I researched the meeting-AI and transcription industry and wrote the field review section of the technical report. I covered the major existing tools — Otter.ai, Fireflies.ai, Zoom AI Companion, Fathom, and tl;dv — and analysed what they offer and what they consistently lack. Most produce a single undifferentiated list of bullet points; none separate decisions from notes. I cited sources from ACL Anthology, arXiv, and industry reports to ground the review in published research.

Labeled Transcripts

I contributed approximately 8 labeled transcripts to `eval_set.jsonl`, including excerpts from public city council meeting minutes — real, verbatim, and messy, with decisions both with and without clear owners.

What I Learned

Doing the field review made me understand the difference between what NLP research has achieved and what is actually deployed in real products. There is a large body of work on meeting summarization, action item extraction, and dialogue act classification going back years, yet mainstream commercial tools still produce simple undifferentiated bullet lists. The research-to-product gap is real and wider than I expected. I also developed a clearer understanding of the different NLP subtasks our system combines: automatic speech recognition (Whisper), abstractive summarization (the two-sentence summary), and named entity and relation extraction (pulling out task, owner, and deadline as structured fields). Each has its own failure modes and literature, and the overall system quality is limited by whichever stage performs worst on a given input.